Summary of Scientific Achievements

1. Overview

I am an astrophysicist developing advanced data analysis methods — integrating physics-informed AI, differentiable modeling, and topological data analysis — to **extract fundamental physics from next-generation astronomical surveys**. Since completing my PhD in April 2019 (six months ahead of schedule), I have held postdoctoral positions across three countries (Sweden, Switzerland, and the Netherlands) — including a competitive Nordita fellowship for independent research (2022–2024) — and currently lead an independent research project funded by the Olle Engkvist Foundation at Stockholm University. My work bridges high-dimensional theoretical simulations and complex observational data from facilities such as the Low-Frequency Array (LOFAR), the Square Kilometre Array (SKA), Euclid, and the Nancy Grace Roman Space Telescope, with a **focus on cosmic reionization and the nature of dark matter**.

Over the past seven years, I have authored 73 publications (64 in international peer-reviewed journals, 4 open-source astrophysics codes, and 5 conference proceedings), with an h-index of 29, m-index of 3.2, $i10$-index of 52, and $i100$-index of 5 (Google Scholar, March 2026). I have made significant contributions to more than 80% of these publications, including **15 as first author, 17 as second author, and 8 as third author**. Among the second- and third-author contributions, **15 were led by students and junior researchers whom I supervised or mentored**, reflecting my commitment to developing the next generation of researchers. The evolution of my citation metrics is shown in Figure 1.

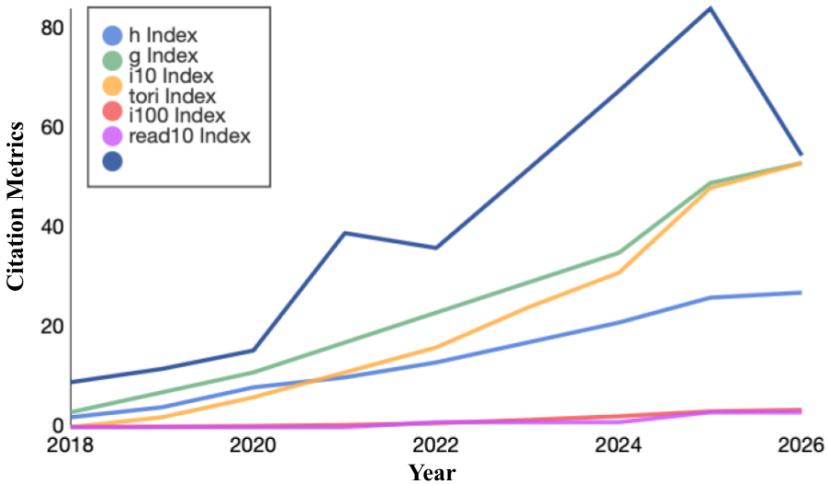

Figure 1: Evolution of citation metrics over the years, retrieved from SAO/NASA Astrophysics Data System (ADS) on 24 March 2026. The most recent data can be found at: SAO/NASA ADS and Google Scholar.

A distinctive feature of my research profile is the breadth of my collaborative network. During my postdoctoral career, I established a substantial body of work with collaborators entirely independent of my PhD supervisor, Prof. Garrelt Mellema (Stockholm University) — most notably with Dr. Aurel Schneider (University of Zurich) on baryonic feedback modeling and dark matter constraints, and more recently with Assoc. Prof. P. Daniel Meerburg (University of Groningen) on primordial non-Gaussianity. This demonstrates my ability to independently initiate new research directions and build productive collaborations across different subfields of cosmology.

2. Doctoral Thesis

I carried out my PhD under the supervision of Prof. Garrelt Mellema at Stockholm University. I defended my thesis, titled "**Tomographic studies of the 21-cm signal during reionization**: Going beyond the power spectrum", in April 2019. My thesis developed state-of-the-art cosmological simulations of the epoch of reionization — the period when photons emitted by the first stars and galaxies ionised the intergalactic medium — and introduced novel image-based statistical techniques to extract astrophysical information from the 21-cm signal. My doctoral thesis comprised four first-author papers and one second-author paper. In addition, I published five further articles during my PhD, including one first-author paper in a different subfield from my thesis supervisor, demonstrating early independence in my research.

3. Research Achievements by Theme

My research programme is organised around four interconnected pillars, each contributing to the overarching goal of extracting fundamental physics from complex cosmological data.

Pillar 1: 21-cm Cosmology and Cosmic Reionization

The 21-cm signal from neutral hydrogen is one of the most promising probes of the early Universe, offering a three-dimensional map of the intergalactic medium during cosmic dawn and the epoch of reionization. A central challenge is that extracting physical information from this signal requires bridging computationally expensive simulations with realistic observational conditions.

I have made foundational contributions to this field across multiple fronts. I developed `Tools21cm`, a widely used open-source Python package for analysing and interpreting 21-cm signal data, which incorporates instrumental effects such as beam convolution, foreground contamination, and thermal noise. `Tools21cm` is extensively used within both the SKA and LOFAR communities and serves as a standard bridge between theoretical simulations and realistic radio-telescope observables.

I pioneered the application of **topological data analysis** to 21-cm cosmology, introducing Betti numbers and percolation statistics as tools for characterising the morphology of ionised regions [**SG10,SG8,SG2**]. These topology-sensitive descriptors capture non-Gaussian structural information that conventional power spectrum analysis misses, providing qualitatively new constraints on the physics of reionization.

To **accelerate the computationally demanding simulations** required for data interpretation, I co-developed `pyC`^{2}`Ray` (a GPU-accelerated radiative transfer framework) and `BEoRN` (a fast and flexible simulation code optimised for exploring the high-dimensional parameter space of reionization). Both tools enable rapid forward modeling, and BEoRN in particular can be directly coupled to statistical inference pipelines.

I developed a **Bayesian inference framework** using Gaussian-process emulators to interpret the Low-Frequency Array (LOFAR) epoch of reionization observations [**SG9**], which provided upper limits on the 21-cm power spectrum. This demonstrated that machine-learning surrogate models can make otherwise intractable simulation-based inference practical. Looking ahead, the upcoming SKA will deliver direct detections of the 21-cm signal with sufficient sensitivity to go beyond the power spectrum and exploit the rich non-Gaussian information encoded in tomographic images. To prepare for this, I have extended the interpretation strategy to simulation-based inference (SBI) using neural density estimators trained on forward-modeled datasets [**SG1**], enabling likelihood-free posterior estimation that integrates multiple summary statistics — including power spectra, higher-order correlations, and topological descriptors — within a unified Bayesian framework.

Most recently, I began exploring Fourier phase statistics as a novel probe of non-Gaussian information in the 21-cm signal (Giri & Meerburg, in prep.). The 21-cm signal traces matter evolution during cosmic dawn and encodes signatures of primordial physics. However, standard non-Gaussianity metrics such as the bispectrum require very high signal-to-noise ratios to yield useful constraints [**SG1**] and face the fundamental challenge of disentangling astrophysical non-Gaussianity — generated by reionization processes [**SG8**] — from the primordial non-Gaussianity of interest. Fourier phase statistics offer a complementary approach that is more robust to these contaminations, opening a new window into primordial physics through next-generation radio observations.

Pillar 2: Dark Matter Constraints from Cosmological Observations

Understanding the nature of dark matter is one of the central questions in modern physics. Different dark matter models — warm, mixed, decaying, or hot — leave distinct imprints on the small-scale matter distribution, which propagate into observable signatures across cosmic time. I demonstrated that 21-cm observations can constrain warm dark matter models by exploiting the morphology of ionised regions at high redshift [**SG5**], showing that SKA-sensitivity data can significantly improve on existing bounds. I extended this to mixed dark matter scenarios involving both fermionic and bosonic components [**SG5**], and showed that JWST galaxy data provide complementary constraints through early galaxy formation [**SG3**].

In the late-time Universe, I contributed to constraining decaying dark matter using cosmic microwave background and weak lensing data and hot dark matter sub-species using similar multi-probe approaches [**e.g., SG4**]. These works explore whether non-standard dark matter scenarios can alleviate the S_8 tension between weak lensing and CMB measurements — a key open question in contemporary cosmology.

My dark matter research thus spans both high-redshift and low-redshift probes, using a unified methodological approach based on forward modeling and machine-learning emulation to constrain particle physics from cosmological structure.

Pillar 3: Baryonic Feedback Modeling for Precision Cosmology

Baryonic feedback processes — such as AGN heating and stellar outflows — redistribute matter on megaparsec scales, suppressing clustering in ways that can mimic or obscure fundamental-physics signals in weak lensing surveys. Precision cosmology with Euclid and other Stage-IV surveys requires controlled modeling of these effects.

I developed `BCemu` [**SG7**], a machine-learning emulator that predicts the suppression of the matter power spectrum due to baryonic feedback. `BCemu` is built on the baryonification framework, which modifies dark-matter-only N-body simulations by redistributing mass within halos according to parametrised gas and stellar profiles. `BCemu` reproduces the results of expensive hydrodynamical simulations (including Illustris-TNG, EAGLE, and BAHAMAS) at sub-percent accuracy and has been **integrated into the official Euclid mission end-to-end pipeline for constraining baryonic physics and dark matter models**.

Using `BCemu`, we performed joint analyses combining weak-lensing measurements with X-ray cluster observations and kinetic Sunyaev–Zel'dovich data [**SG6**], demonstrating how astrophysical feedback and cosmological parameters can be disentangled within a unified inference framework. I have since extended `BCemu` into **a differentiable, JAX-based emulator** trained on an improved baryonification framework, enabling gradient-based inference in high-dimensional parameter spaces for upcoming Stage-IV survey analyses.

Pillar 4: Open-Source Tools and Methods Infrastructure

I am a strong advocate of open science and have developed several widely used software packages that serve both the research community and educational purposes:

- `Tools21cm`: Analysis and interpretation of 21-cm signal data; standard tool in SKA and LOFAR communities.
- `pyC`^{2}`Ray`: GPU-accelerated radiative transfer for cosmic reionization simulations.
- `BEoRN`: Fast reionization simulation code for parameter space exploration and inference.
- `BCemu`: Baryonic correction emulator; integrated into the Euclid pipeline.
- `AstronomyCalc`: A pedagogical Python toolkit for university-level teaching of complex and abstract concepts in cosmology and data analysis.
- `StoReS`: Public data repository for reionization simulations.

These tools reflect my commitment to reproducible, community-driven science and have collectively enabled research across multiple international collaborations.

4. Grants and Funding

I have successfully secured funding from multiple agencies, demonstrating that my research programme attracts support from diverse sources. Below I list the principal grants in reverse chronological order.

Research grants:

- Olle Engkvists Stiftelse (PI), 2024: 1.8M SEK (∼170k EUR) for the research project *Cosmic Reionization and Dark Matter: Unveiling the Connection.*
- Nordita PhD Fellows visitor program (PI), 2024: In-kind support to host PhD student Timothée Schaeffer (University of Zurich) for 1.5 months.
- Wenner-Gren Foundations (co-I), 2023: 150k SEK (∼15k EUR) for organising the Cosmic Dawn at high latitudes conference.
- Royal Swedish Academy (co-I), 2023: 150k SEK (∼15k EUR) for organising the Cosmic Dawn at high latitudes conference.
- Nordita PhD Fellows visitor program (PI), 2023: In-kind support to host PhD student Anshuman Acharya (Max Planck Institute for Astrophysics) for 2 months.
- Nordita workshop (co-PI), 2023: 600k SEK (∼57k EUR) for organising a 4-week workshop, *Cosmic Dawn at high latitudes.*
- Swedish Research Council, VR Project grant (co-I; PI: Prof. Garrelt Mellema), 2021: 3.12M SEK (∼0.3M EUR) for 4 years, *Explorations of the Epoch of Reionization.*

High-performance computing grants (selected):
I have been awarded substantial computing allocations essential for carrying out large-scale simulations for current and upcoming telescopes, including over 60 million core hours on Swedish national facilities (Dardel, Tetralith) via NAISS/SNIC as co-PI (2023–2024), 70,000 node hours at the Swiss National Supercomputing Centre (2022), and GPU allocations on LUMI in Finland (2024). Earlier allocations as named collaborator supported my PhD-era simulation work on Beskow.

Earlier career grants:
During my PhD and early postdoc, I received 14 smaller grants (travel, equipment, conference attendance) from sources including the Knut and Alice Wallenberg Foundation, IAU, University of Zurich GRC, and Stockholm University, totaling ∼100k SEK. These early grants supported the international network-building that underpins my current collaborative programme.

5. Awards and Distinctions

My research contributions have been recognised through several competitive awards and fellowships at both international and national levels.

- **Young Scientist Award**, International Union for Radio Science (URSI), 2023
- **Nordita Fellowship** for independent research, 2022–2024
- **Gold Medal** (equivalent to summa cum laude), Integrated Master's degree, IIT (BHU), 2015
- **National Talent Search Examination** (NTSE) scholarship, Government of India, 2008 — a nationwide competitive programme established to identify talented students and support them in pursuing higher education in the sciences, selecting approximately 1,000 scholars in India annually.

6. Peer Review Activities

I serve as a regular reviewer for leading journals in the field, reflecting broad expertise across astrophysics and cosmology:

- Since 2020: Journal of Open Source Software, Monthly Notices of the Royal Astronomical Society
- Since 2022: Journal of Cosmology and Astroparticle Physics
- Since 2023: Physical Review Letters
- Since 2024: Astronomy & Astrophysics, Nature Communications Physics, Physical Review D
- Since 2025: The Astrophysical Journal

7. Scientific Publications

In total, I have authored 73 publications, including 64 in international peer-reviewed journals, 4 open-source astrophysics codes, and 5 conference proceedings. I have made significant contributions to more than 80% of these publications, including 15 as first author, 17 as second author, and 8 as third author. Among the second- and third-author contributions, 15 were led by students and junior researchers whom I supervised or mentored. Publications that formed part of my doctoral thesis are marked with (†).

Doctoral Thesis

Giri S. K., 2019. Tomographic studies of the 21-cm signal during reionization: Going beyond the power spectrum. PhD thesis, Stockholm University.

Articles in International Peer-Reviewed Journals

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Upcoming

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